

Data Link Layer:

- Sends data from Hop to adjacent Hop (Nodes to Nodes) using the MAC address.
- This layer is a combination of hardware, software, firmware.
- Uses ARP to determine MAC addresses
- Implemented at the NIC (Network Interface Card) adaptor in the computer

Functions:

- Error detection & correction
- Sharing a broadcast channel both half and full duplex
- Framing: adds header (MAC address included) and trailer

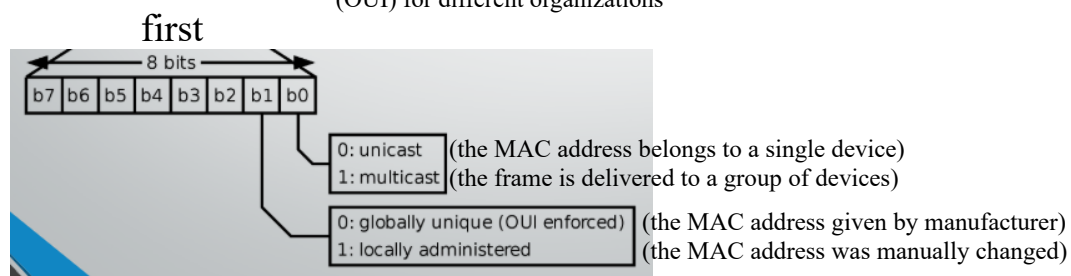
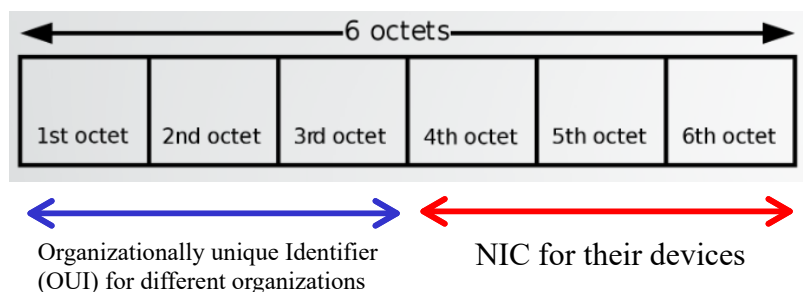
MAC Address:

48 bits unique identifier assigned to a NIC

12 HEX digits (6 pairs, divided by colons or hyphens)

Portable (stays the same no matter what network you're in)

Assigned by IEEE



MAC Address when:

1. Unicasting:

Source MAC: Host MAC, Destination MAC: Receiver MAC

2. Broadcasting:

Source MAC: Host MAC, Destination MAC: FF-FF-FF-FF-FF-FF

3. Multicasting:

Source MAC: Host MAC, Destination MAC: 01-00-5E-XX-XX-XX

ARP: Address Resolution Protocol

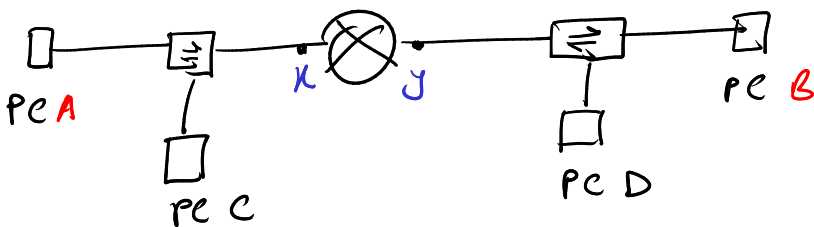
Used to determine MAC addresses of devices when IP is known.
Creates an ARP table to map IP-MAC but has a TTL of typically 20min

Steps when both devices in the same network:

- A wants to know the MAC of B
- A Broadcasts ARP query packet containing B's IP address
- All except B drops the packet
- B responds with their own MAC address (unicast)

FF-FF-FF-FF-FF-FF	71-65-F7-2B-08-53	137.196.7.14	137.196.7.23	ARP Request
Dest MAC	Source MAC	Dest IP Add	Source IP Add	ARP Req/Reply

71-65-F7-2B-08-53	58-23-D7-FA-20-B0	137.196.7.23	137.196.7.14	ARP Reply
Dest MAC	Source MAC	Dest IP Add	Source IP Add	ARP Req/Reply



A, B, x, y all have both IP & MAC addresses of their own

When two devices in different networks:

- A first has to broadcast ARP request to the default gateway of the router and the router returns its own MAC address. Since A cannot broadcast to another network, router because the router will drop the packet.
- A then sends the packet to the router with the MAC address of the default gateway (was broadcast before) and the router sees the destination address is of B.
- Router now sends an ARP to find the MAC of B.
- After it receives the MAC address of B, it sends the packet to B

MAC src: A
MAC dst: broadcast
IP src: A
IP dst: B

MAC src: A
MAC dst: Router x
IP src: A
IP dst: B

MAC src: Router y
MAC dst: broadcast
IP src: A
IP dst: B

MAC src: Router y
MAC dst: B
IP src: A
IP dst: B

(A never gets the MAC address of B, it gets the MAC address of the router gateway using ARP and then sends the intended packet to the router, the router then gets the MAC address of B using ARP and sends the packet to B)

Switches:

- Self learning
- Records MAC address and port number in forwarding tables
- Works only with MAC address
- When target MAC address port is known selectively sends to that port
- When target MAC address port is unknown floods to all ports
- Updates forwarding tables based on the source MAC of Frames